Suggested Itinerary, Route Map & Navigation: UI/UX Evaluation

User & Developer Perspectives • Live Target: edselrenz.github.io/Itinerary-App/tabs/home • Date: November 6, 2026

1. Executive Summary: Post-Save UI Downgrade & Missing Route Visualization

We executed a live end-to-end evaluation on https://edselrenz.github.io/Itinerary-App/tabs/home covering the complete lifecycle of a Suggested Itinerary: generating a new 3-day plan in the Suggested Itinerary Wizard, saving it to local storage, opening the saved trip from Saved Itineraries (/itinerary/f24819d1-4084-4798-8b17-a19e2e87784d), inspecting the route map across views, and testing the turn-by-turn navigation handoff (Open in Google Maps).

Comparing the live user experience against the underlying Angular 22 and Ionic 9 templates (`suggested-wizard.page.html`, `itinerary-detail.page.html`, `history.page.html`, and `map.page.html`) revealed a critical product paradox: **the temporary pre-save preview in the Suggested Wizard is visually richer, more informative, and more interactive than the permanent Saved Itinerary Detail view that travelers actually rely on during their trip.**

- **Complete Absence of Route Maps in Saved Trips:** Neither the Saved Itineraries list nor the Saved Trip Detail view renders an embedded route map, numbered stop markers, or connecting polylines. Switching to the Map Tab displays only an unlinked provincial exploration canvas where marker pins fail to render due to an ES module namespace defect.
- **Severe Visual & Feature Regression After Saving:** Saving a generated plan and opening it in `/itinerary/:id` strips away all stop thumbnail photos, removes the segmented day switcher tabs, truncates time windows from `12:00 PM - 2:00 PM (2 hrs)` to bare `12:00`, hides the *Nearby Places* and *Share / Export* triggers (leaving orphaned dead code in the template), and removes the hero photo, close button, opening hours, fare, and drop-off point from the Stop Detail modal.
- **High-Friction Single-Stop Navigation & Contradictory Cost Math:** Clicking **Open in Google Maps** inside a stop modal only routes from the user's live location to that single stop (?api=1&destination=15.6315,121.144) rather than chaining the day's itinerary stops as waypoints. Meanwhile, `itinerary-detail.page.ts:580` fabricates a hardcoded `₱200` fee for every unpriced venue, causing the bottom total card to contradict the top `Verified Estimated Costs` banner on the same screen.

2. Step-by-Step Live Walkthrough & Behavioral Trace

Each step of the user journey was executed in a live browser session and cross-referenced against the exact component handlers in `Project/src/app/`:

1. **Launching the Generator (/tabs/home → /suggested-wizard):** Tapping the **Suggested Itinerary** card opens the questionnaire form with `Destination = "Nueva Ecija"`, `Travel Dates = "Nov 6 – 8, 2026"` (3 days, 2 nights), `Daily Budget = 3000`, and pre-selected preference pills (*Food Trip*, *Culture & History*, *Resort*).
2. **Generating & Reviewing the Preview (/suggested-wizard, `viewMode === 'itinerary'`):** Tapping **Find Itinerary** displays a spinner (*"Clustering recommendations…"*) and renders a 3-day plan: *Day 1: Bongabon* (Rohi Paperflower Cafe, Museo de Bongabon, Thornes Samgyupsal House, La Casa Prime Cabins), *Day 2: Cabanatuan City / San Jose*, and *Day 3: Gapan City / General Tinio*. The preview includes a top summary card with thumbnail, a green `Verified Estimated Costs` ledger (`₱3,449 - ₱8,349 (+ 3 unpriced)`), collapsible color-coded `Trip Assumptions & Travel Notes` cards, a segmented day capsule bar, `68×68px` photo stop cards, and a dashed blue **Nearby Places in Bongabon** trigger banner.
3. **Saving & Opening from Saved Itineraries (/tabs/history → /itinerary/f24819d1-4084-4798-8b17-a19e2e87784d):** Tapping **Accept & Save Itinerary** saves the trip to IndexedDB/localStorage, shows a success toast, and adds the card to `/tabs/history`. Tapping the saved card opens `ItineraryDetailPage`.
4. **Looking for the Route Map & Testing Navigation:** On `/itinerary/f24819d1-4084-4798-8b17-a19e2e87784d`, there is **no route map** and **no button to view the itinerary on the in-app map**. Tapping Stop 1 (*Rohi Paperflower Cafe*) opens a minimal modal sheet with a shortened *"How to get there?"* block (`By Bus / UV Express • Cabanatuan City Central`

`Terminal`, `1 hr – 1 hr 30 mins`) and an **Open in Google Maps** button that opens https://www.google.com/maps/dir/?api=1&destination=15.6315,121.144 in a new browser tab.

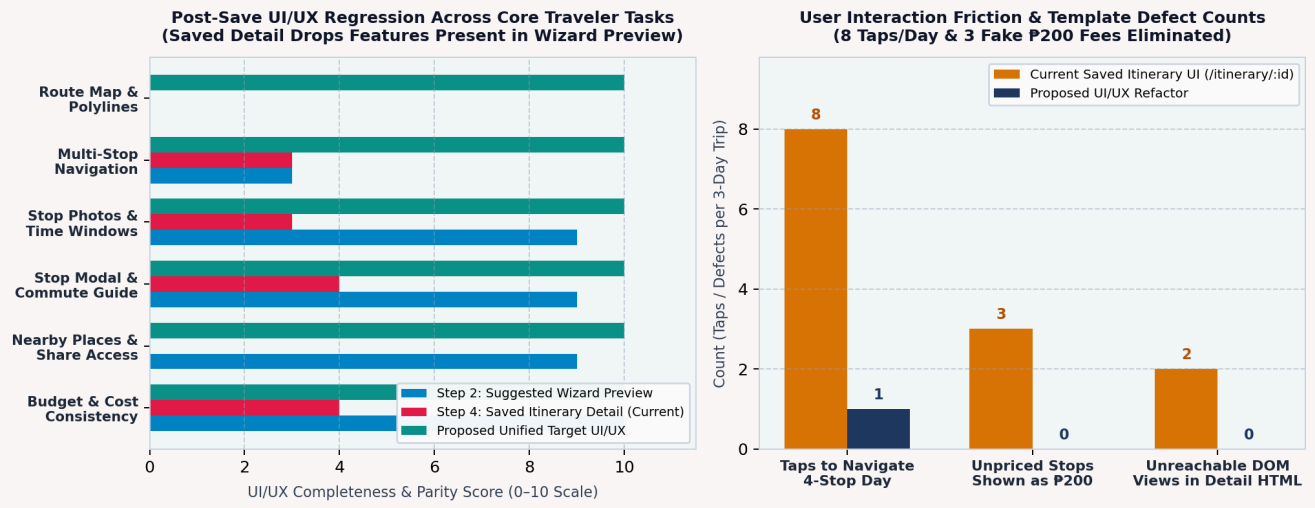


Figure 1: Left: UI/UX Feature & Visual Completeness audit across 6 core traveler tasks, illustrating the sharp drop from the Suggested Wizard Preview (`/suggested-wizard`) to the Saved Itinerary Detail screen (`/itinerary/:id`). Right: Navigation interaction friction (8 taps required to navigate 4 stops individually vs. 1 tap with a multi-waypoint day route launcher) and the ₱600 artificial cost inflation caused by `itinerary-detail.page.ts:580`.

3. Screen-by-Screen UI/UX Parity & Regression Matrix

The matrix below contrasts how identical trip data is rendered in the Suggested Wizard preview versus the Saved Itinerary Detail screen, pinpointing the exact template files responsible for each discrepancy.

UI / UX Dimension	Suggested Wizard Preview (`suggested-wizard.page.html`)	Saved Itinerary Detail (`itinerary-detail.page.html`)	Proposed Unified Design (User + Developer Target)
1. Route Map & Polylines	None. Displays vertical stop cards only without spatial preview.	None. No embedded map, no stop pins, and no deep-link to /tabs/map.	Collapsible `220px` interactive Leaflet route map with numbered pins (`1 → 2 → 3`), OSRM/geodesic polylines, and **Expand on Map Tab** CTA.
2. Turn-by-Turn Navigation	Single-stop external link inside modal (`openInGoogleMaps(stop)`).	Single-stop link (?destination=15.6315,121.144); ignores stop sequence.	Primary **Navigate Day Route** button (`&origin=...&destination=...&waypoints=...`) plus inline per-stop navigate icon buttons.
3. Day Switching vs. Scrolling	Segmented capsule bar (`.figma-day-capsule-bar`, L221–233) focusing on 1 day at a time.	Dumps all days into one long vertical list (`.days-view-list`, L175–207); `selectDay()` in TS is unused.	Sticky segmented day switcher capsule bar synchronized with the active day's route map and timeline cards.
4. Stop Card Visuals & Photos	Rich `.figma-stop-card` (L244–260) with 68×68px photo thumbnail and numbered title.	Plain text-only box (`.activity-view-card`, L185–204) with zero photos and no step numbers.	Shared `<app-itinerary-stop-card>` with photo thumbnail, step badge, category pill, and leg transit connector badge.
5. Time Window Formatting	Full range + duration via `getTimingTime(stop)`: `12:00 PM - 2:00 PM (2 hrs)`.	Calls `formatTimingHour(stop)` (L188), stripping times to bare `12:00` without AM/PM or end time.	Bind `getTimingTime(stop, sIdx)` consistently so every stop displays its complete start–end window and duration.

UI / UX Dimension	Suggested Wizard Preview (`suggested-wizard.page.html`)	Saved Itinerary Detail (`itinerary-detail.page.html`)	Proposed Unified Design (User + Developer Target)
6. Cost Honesty & Rollup	Honest range via `budgetSummary.displayText`: `₱3,449 - ₱8,349 (+ 3 unpriced)`.	`getStopEstimatedCost()` (TS L580) returns hardcoded `'200'` for `null` costs, inflating bottom total by ₱600.	Never fabricate `₱200` for unverified stops; show `Fee not listed` on unpriced cards and use `budgetSummary` everywhere.
7. Stop Detail Modal	Full modal (L392–495): `240px` hero image, X close button, category badge, hours, and cost pill.	Stripped modal (L367–423): no photo, no X close button, no hours, and no cost pill.	Extract a single shared `<app-stop-detail-modal>` used across Home, Map, Wizard, and Itinerary Detail.
8. Commute Guide (*How to get there?*)	Shows all 4 fields (`Transit & Hub`, `Drop-off Point`, `Travel Time`, `Fare`) + `Travel Reminder`.	Omits `Drop-off Point`, `Estimated Fare`, and `Travel Reminder` (L389–408); always assumes Cabanatuan hub.	Render all 4 fields + reminder, and compute origin-aware leg advice (walk/tricycle for intra-town hops vs. inter-town transit).
9. Nearby Essentials Explorer	Interactive dashed banner (`.nearby-banner-trigger`, L263–273) opening 3-tab essential services view.	**Unreachable dead view:** `viewMode === 'nearby'` exists at L62–112, but no button in HTML calls `openNearbyPlaces()`.	Restore the **Nearby Places in {{ getDayLocation(day) }}** banner inside each day card and persist added stops via `ItineraryService`.
10. Header Chrome & Share Sheet	Clean single toolbar: `← Suggested Itinerary [Edit]`.	Stacked double header (`Saved Itineraries [+]` above `← Trip Title [Edit]`); Share modal (L428) has no trigger button.	Single unified toolbar (`← Trip Title | [Share] [Edit]`), wiring `openShareModal()` to a visible header action button.

4. User's Point of View: Traveler UX Critique & Suggestions

From the perspective of a traveler planning and executing a multi-day road or commute trip in Nueva Ecija, the current flow creates six concrete usability pain points:

4.1 Eliminate the Post-Save "Bait-and-Switch" Visual Downgrade

- **What the User Experiences:** During step 2 of the Suggested Itinerary Wizard, the itinerary looks like a modern travel companion with vibrant photography, day-by-day tabs, clear time ranges, and nearby dining/pharmacy lookup. Once the user clicks **Accept & Save Itinerary** and opens the saved trip at /itinerary/f24819d1-4084-4798-8b17-a19e2e87784d, the interface turns into a plain text ledger with no photos, truncated timestamps (`12:00`), and missing tools.
- **UX Suggestion:** Make the Saved Itinerary Detail screen the visual flagship of the app. Retain the destination thumbnail on every stop card, preserve the full `12:00 PM - 2:00 PM (2 hrs)` schedule badge, and keep the segmented `Day 1 | Day 2 | Day 3` capsule bar sticky below the header so travelers can switch days with one thumb tap.

4.2 Embed an Interactive Day Route Map & Transit Leg Connectors

- **What the User Experiences:** Travelers looking at their saved itinerary have no visual sense of direction, distance, or sequence across Nueva Ecija. Worse, switching to the Map Tab shows an empty provincial map that does not display their saved trip.
- **UX Suggestion:** Place a collapsible `220px` route map card directly above the active day's stop timeline showing numbered markers (`1 → 2 → 3 → 4`) connected by a route line, paired with a **View Fullscreen on Map Tab** button. Between each consecutive stop card in the vertical timeline, insert a compact transit connector spine (e.g., `↓ 1.2 km • ~4 mins drive / tricycle`) so the user immediately sees the travel gap between activities.

4.3 Replace 8-Tap Single-Stop Handoffs with One-Tap Multi-Stop Day Navigation

- **What the User Experiences:** There is no *Navigate* button on the itinerary detail page or on the stop cards themselves. To navigate a 4-stop day, the user must tap Stop 1 → scroll the modal → tap Open in Google Maps → switch back to the browser → dismiss the modal → tap Stop 2 → repeat (8+ taps total), and Google Maps only routes from the user's current GPS position to one stop at a time.
- **UX Suggestion:** Add a prominent primary action bar on each day header featuring **Start Day Route in Google Maps** (which opens a single multi-waypoint Google Maps route for all stops on that day in sequence), and place a quick-action directions icon button (`navigate-outline`) on the right edge of every stop card for one-tap single-leg navigation without opening the detail modal.

4.4 Make the "How to Get There?" Commute Guide Leg-Aware

- **What the User Experiences:** On Day 1 in Bongabon, Stop 1 is *Rohi Paperflower Cafe* and Stop 2 is *Museo de Bongabon* (located just 200 meters away in the same town proper). Yet tapping Stop 2 tells the user: *"Transit & Hub: By Bus / UV Express • Cabanatuan City Central Terminal • 1 hr – 1 hr 30 mins."* Furthermore, on the Saved Trip screen, the modal cuts off the *Drop-off Point*, *Estimated Fare* (`₱66 – ₱80`), and *Travel Reminder* that were visible in the Wizard!
- **UX Suggestion:** Restore all four commute fields and the reminder callout in the Saved Trip modal, and make the commute guide aware of the previous stop: display the inter-city bus/UV Express terminal instructions on **Stop 1 of the day** (or when crossing between municipalities), and display local transfer instructions (*"From Rohi Paperflower Cafe (0.2 km): 3 min walk or ₱15–₱20 local tricycle"*) for intra-municipality stops.

4.5 Clean Up Confusing Header Chrome, Raw ISO Dates & False Affordances

- **What the User Experiences:** The Saved Trip screen stacks two headers on top of each other: `Saved Itineraries [+]` and `← Suggested Nueva Ecija Trip [Edit]`. Tapping `[+]` unexpectedly abandons the saved trip and opens the Custom Builder. Inside the `Trip Details` card, dates render as raw database strings with wide gaps (`2026-11-06  2026-11-08`), and a right arrow (→) sits in the card footer looking like a clickable button that does nothing when tapped.
- **UX Suggestion:** Remove the redundant top `Saved Itineraries [+]` bar on `/itinerary/:id`, format dates human-readably via `formatDateRange()` (`Nov 6 – Nov 8, 2026 (3 Days, 2 Nights)`), remove the non-interactive → icon, and place **Share/Export** (`share-social-outline`) and **Edit** (`create-outline`) side by side in the single header bar.

5. Developer's Point of View: Codebase, Template & Architecture Critique

From a software engineering and Angular architecture perspective, the UX anomalies observed in live testing stem from six specific code defects in `Project/src/app/`:

5.1 Orphaned Template Views & Unbound Component Methods in `itinerary-detail.page.html`

Inspecting `Project/src/app/features/itinerary/itinerary-detail/itinerary-detail.page.ts` alongside `itinerary-detail.page.html` reveals that Phase 23 replaced the View Mode markup (lines 113–216) to match a simplified static screenshot, accidentally severing the template bindings for several fully implemented features:

- **Unreachable Share & Export Modal:** `<ion-modal class="profile-subwindow-modal share-export-modal" [isOpen]="isShareModalOpen">` is rendered at lines 428–452 of `itinerary-detail.page.html`, and `openShareModal()`, `exportIcs()`, `copySummary()`, `shareNative()`, and `printPdf()` are implemented at lines 844–896 of `itinerary-detail.page.ts`—**but zero buttons in `itinerary-detail.page.html` call `openShareModal()`.**
- **Unreachable Nearby Places View:** `@if (viewMode === 'nearby')` is rendered at lines 62–112 of `itinerary-detail.page.html`, and `openNearbyPlaces()` is defined at line 616 of `itinerary-detail.page.ts`—**but zero buttons in `itinerary-detail.page.html` call `openNearbyPlaces()`.**
- **Unused Day Selection & Rich Time Formatters:** `activeDayIndex`, `selectDay(index)` (line 605), `formattedDateSpan` (line 645), and `getTimingTime(stop, index)` (line 700) remain in `itinerary-detail.page.ts` but are bypassed by the template in favor of raw `{{ itinerary.start_date }}` and `formatTimingHour(stop)`.

5.2 Hardcoded `?? 200` Cost Fallbacks Violating Honest Budget Accounting (`ADR-001`)

In `itinerary-detail.page.ts` (lines `575–603`), two helper methods introduce synthetic cost numbers whenever a stop has `estimated_cost_min = null`:

```
// Project/src/app/features/itinerary/itinerary-detail/itinerary-detail.page.ts (Lines 575–603)
public getStopEstimatedCost(item: ItineraryItem): number | string {
  const cost = item.estimated_cost_min ?? item.estimated_cost_max;
  if (cost != null && cost > 0) {
    return cost.toLocaleString();
  }
  return '200'; // BUG: Fabricates ₱200 for free (₱0) and unverified (null) venues!
}

public calculateTotalActivityExpenses(): number {
  if (!this.itinerary?.days) return 0;
  let sum = 0;
  for (const day of this.itinerary.days) {
    for (const item of day.items || []) {
      const cost = item.estimated_cost_min ?? item.estimated_cost_max ?? 200; // BUG: Adds ₱200 per unpriced
stop
      if (cost && !isNaN(Number(cost))) sum += Number(cost);
    }
  }
  return sum;
}
```

In our 3-day test trip, three stops (*Museo de Bongabon*, *Pantabangan Dam Viewpoint*, and *Tagpuan sa Lumang Gapan*) have `null` costs in `offline-places.data.ts`. Because of `?? 200`, each card falsely displays `Est. ₱200`, and the bottom `Estimated activity expenses` card displays `₱4,049` while the top `Verified Estimated Costs` banner displays `₱3,449 - ₱8,349 (+ 3 unpriced)`. Both methods should delegate directly to `getStopCost(item)` (line `759`) and `this.budgetSummary`.

5.3 Destructive Regex in `formatTimingHour()` & Hidden Custom Assumptions

- **Regex Time Truncation:** `formatTimingHour(item)` (`itinerary-detail.page.ts:557–565`) executes `item.planned_time.match(/(\d{1,2}:\d{2})/)` and returns `match[1]`. When `planned_time` is `"12:00 PM - 2:00 PM"`, it strips out the meridian and end time, returning `"12:00"`; when `planned_time` is `"1 Night · 2 Days"` (for *La Casa Prime Cabins*), the regex fails and returns the raw string above `"22 hours"`. Binding `getTimingTime(stop, sIdx)` fixes this immediately.
- **Hidden Budget & Assumptions Cards on Custom Plans:** In `itinerary-detail.page.html:149, 160`, `[class.hidden-on-custom]="itinerary.type !== 'suggested'"` applies `display: none !important` to `.verified-budget-card` and `.assumptions-info-box` whenever a user opens a Custom Itinerary—hiding verified budget warnings on custom trips.

5.4 Multi-Waypoint URL Builder & Shared Route Map Component Blueprint

To unify navigation across `SuggestedWizardPage`, `CustomBuilderPage`, `ItineraryDetailPage`, and `MapPage` without violating `ADR-001` (Zero Paid APIs) or Angular's `10 kB` per-component SCSS budget (`ADR-009`):

```
// Project/src/app/core/utils/itinerary-export.util.ts — Multi-Stop Day Route URL Builder
export function buildGoogleMapsDayRouteUrl(items: ItineraryItem[]): string | null {
  const coords = (items || [])
    .map((it) => ({
      lat: it.place?.latitude ?? it.custom_lat,
      lng: it.place?.longitude ?? it.custom_lng,
    }))
    .filter((c): c is { lat: number; lng: number } => typeof c.lat === 'number' && typeof c.lng === 'number');

  if (coords.length === 0) return null;
  if (coords.length === 1) {
    return `https://www.google.com/maps/dir/?
api=1&destination=${coords[0].lat},${coords[0].lng}&travelmode=driving`;
  }
  const origin = `${coords[0].lat},${coords[0].lng}`;
  const destination = `${coords[coords.length - 1].lat},${coords[coords.length - 1].lng}`;
  const middle = coords.slice(1, -1).map((c) => `${c.lat},${c.lng}`).join('|');
```

```
  const waypointsParam = middle ? `&waypoints=${encodeURIComponent(middle)}` : '';
  return `https://www.google.com/maps/dir/?
api=1&origin=${origin}&destination=${destination}${waypointsParam}&travelmode=driving`;
}
```

5.5 prioritized Developer Action Checklist

Priority	Target Component / File	Concrete Code & UI/UX Change	User & Dev Impact
P0 Immediate	`itinerary-detail.page.html` & `.ts`	Restore header Share button (`openShareModal()`) and per-day Nearby banner (`openNearbyPlaces()`); replace `formatTimingHour()` with `getTimingTime()`; remove `?? 200` cost fallback; format dates via `formattedDateSpan`; remove dead → icon.	Eliminates post-save feature loss, fake ₱200 fees, and raw ISO date strings in under 40 lines of template changes.
P1 High	`itinerary-export.util.ts`; `itinerary-route-map.component.ts`	Add `buildGoogleMapsDayRouteUrl(day.items)` and embed a reusable collapsible Leaflet route map with numbered stop pins, geodesic/OSRM polylines, and leg distance/time connector pills.	Reduces 4-stop navigation from 8 taps to 1 tap and gives travelers visual route context inside Saved Itineraries.
P2 Medium	`stop-detail-modal.component.ts`; `commute-guide.data.ts`	Unify the Stop Detail Modal across Wizard and Detail views (restoring hero photo, X close button per WCAG 2.2, hours, fare, drop-off point, and reminder) and make commute guidance origin-aware.	Eliminates 150+ lines of duplicated modal HTML/SCSS and stops advising inter-city buses for 200m same-town hops.